No finite homogeneous bound for the stated magnetic Neumann strip problem

Candidate counterexample packet for arXiv:2106.08625

13 August 2026

Status. Candidate counterexample and full negative answer to the optimal-constant question in Chapter 4 of Sorowen [1], pending expert review. For the standard magnetic Neumann realization on the strip printed in the source, the optimal homogeneous constant is infinite: the underlying estimate already fails.

1 Source estimate and open question

The source is B. Sorowen, *Estimation of the number of negative eigenvalues of magnetic Schrödinger operators in a strip*, arXiv:2106.08625v1 [1]. On PDF page 40 (printed page 31), Theorem 3.2.1 asserts the following estimate on $S = \mathbb{R} \times (0, d)$:

$$N_-(H_{B,V}) \leq \left({\sum_{k\in\mathbb{Z}}}' \frac{1}{\sqrt{16|k+\varphi|^2+1}} \right) \|V\|_X, \qquad X = L^1(\mathbb{R}; L^\infty(0,d)).$$

The prime is described as neglecting summands below one.

3.2 The main result

Theorem 3.2.1 (cf.Theorem 2.5.3). *Let $X = L^1(\mathbb{R}, L^\infty(0,d))$, A be the vector potential in (3.1.2), V be integrable on bounded subsets of S and $V \in X$. Then $Neg(H_{B,V})$ is bounded above by*

$$Neg(H_{B,V}) \leqslant {\sum}'_{k\in\mathbb{Z}} \frac{1}{\sqrt{16|k+\varphi|^2+1}} \|V\|_X, \tag{3.2.1}$$

where ${\sum}'$ indicates that all summands less than 1 are neglected.

On PDF page 42 (printed page 33), Section 4.2 asks for the optimal constant in this estimate.

In this study, the spectral estimate of the magnetic Schrödinger operator with Aharonov-Bohm magnetic field in a straight strip subject to Neumann boundary conditions has been obtained. In particular, we have obtained an upper estimate (3.2.1) for the number of negative eigenvalues for the operator (3.1.1). We have shown in §(3.3) that the upper estimate (3.2.1) depends on the properties of the magnetic field B and that such an estimate fails in the absence of magnetic field and only holds in the presence of the Aharonov-Bohm magnetic field with the flux satisfying the non integer condition.

4.2 Recommendations

There are several directions in which one can extend the above results, for instance, treating the problem using Numerical techniques, one can investigate whether mixed (Dirichlet, Neumann and Robin) boundary conditions applied on the strip can generate similar results.

An open problem here concerns the optimal constant (what is the best constant?) in the estimate. In an attempt to answer this question, one has two options; one is either treating it analytically or using Numerical techniques.

Also, one can extend the above results to a curved quantum waveguide with magnetic fields, where in this case the curvature also contributes to the electric potential.

Finally, instead of restricting the magnetic field to non-integer flux values, it would be interesting to consider a reasonably large class of fields by removing the singularities at integer points.

The natural precise interpretation of that question is whether there is a finite $C = C(d, \varphi)$ such that

$$N_-(H^N_{A_\varphi,V}) \le C\|V\|_X \quad \text{for every nonnegative } V \in X, \tag{1}$$

where H^N denotes the standard gauge-covariant magnetic Neumann realization. The result below disproves (1) for every φ, including the noninteger values singled out in the source.

2 Definitions

Put

$$S = \mathbb{R} \times (0, d), \qquad \theta(x, y) = \operatorname{Arg}(x + iy) \in (0, \pi),$$

and use the source's Aharonov–Bohm potential

$$A_\varphi(x, y) = \varphi \left(-\frac{y}{x^2 + y^2}, \frac{x}{x^2 + y^2} \right) = \varphi \nabla \theta(x, y). \tag{2}$$

The pole $(0, 0)$ is not in S: it lies on its boundary. Consequently, θ is a globally single-valued smooth function on S.

The standard magnetic Neumann operator is associated with the form

$$q_{A_\varphi, V}[u] = \int_S |(-i\nabla + A_\varphi)u|^2 - \int_S V|u|^2, \tag{3}$$

with the gauge-transformed Neumann form domain. Equivalently, if

$$(U_\varphi f)(x, y) = e^{-i\varphi\theta(x,y)} f(x, y),$$

then U_φ is unitary and

$$(-i\nabla + A_\varphi)U_\varphi f = U_\varphi(-i\nabla f), \qquad H^N_{A_\varphi, V} = U_\varphi(-\Delta_N - V)U_\varphi^{-1}. \tag{4}$$

This is the gauge-covariant magnetic Neumann convention used in the waveguide literature; compare Borisov–Ekholm–Kovařík [2].

3 Proof intuition

An Aharonov–Bohm flux has a spectral effect only when the accessible domain retains nontrivial topology around its pole. Here the pole is on the boundary of an open, simply connected strip. Formula (4) therefore removes the potential completely. The Neumann strip has a zero-energy transverse mode, so along that mode the problem is one-dimensional. A long tent function costs only $2/L$ in kinetic energy while a unit interval of attractive potential gains ε. Taking $L > 2/\varepsilon$ produces a bound state at arbitrarily small X-norm, which is incompatible with any homogeneous counting bound.

4 Counterexample

Theorem 1 (No finite best constant)**.** *Let $d > 0$ and $\varphi \in \mathbb{R}$. For every $\varepsilon > 0$ there is a bounded, compactly supported, nonnegative $V_\varepsilon \in L^1(\mathbb{R}; L^\infty(0, d))$ such that*

$$\|V_\varepsilon\|_X = \varepsilon, \qquad N_-(H^N_{A_\varphi, V_\varepsilon}) \geq 1.$$

Consequently, no finite constant $C(d, \varphi)$ can satisfy (1); the optimal homogeneous constant asked for in the source is infinite (equivalently, it does not exist as a finite constant).

Proof. Fix $\varepsilon > 0$ and choose $L > 2/\varepsilon$. Let

$$I_L = [L + 1, L + 2], \qquad V_\varepsilon(x, y) = \varepsilon \mathbf{1}_{I_L}(x).$$

Since I_L has length one,

$$\|V_\varepsilon\|_X = \int_{\mathbb{R}} \operatorname*{ess\,sup}_{0<y<d} V_\varepsilon(x,y)\,dx = \varepsilon.$$

Define the longitudinal tent function

$$g_L(x) = \begin{cases} (x-1)/L, & 1 \le x \le L+1, \\ 1, & L+1 \le x \le L+2, \\ (2L+2-x)/L, & L+2 \le x \le 2L+2, \\ 0, & \text{otherwise.} \end{cases}$$

It belongs to $H^1(\mathbb{R})$, is supported away from the boundary pole, and obeys

$$\int_{\mathbb{R}} |g_L'(x)|^2\,dx = \frac{2}{L}, \qquad \int_{I_L} |g_L(x)|^2\,dx = 1. \tag{5}$$

Use the constant transverse Neumann mode and undo the gauge:

$$u_L(x,y) = e^{-i\varphi\theta(x,y)}\,\frac{g_L(x)}{\sqrt{d}}.$$

By (4) and (5), its exact form value is

$$\begin{aligned} q_{A_\varphi,V_\varepsilon}[u_L] &= \int_{\mathbb{R}} |g_L'|^2 - \varepsilon \int_{I_L} |g_L|^2 \\ &= \frac{2}{L} - \varepsilon < 0. \end{aligned}$$

The unperturbed operator is unitarily equivalent to the Neumann Laplacian on the strip and hence has essential spectrum $[0,\infty)$. The bounded, compactly supported potential is a relatively compact form perturbation, so the essential spectrum remains $[0,\infty)$. The min–max principle now turns the negative trial value into a discrete negative eigenvalue. Thus $N_- \ge 1$.

If a finite C in (1) existed, choose $0 < \varepsilon < 1/C$ (with the conclusion immediate when $C = 0$). Then

$$1 \le N_-(H^N_{A_\varphi,V_\varepsilon}) \le C\varepsilon < 1,$$

a contradiction. □

5 Verification and diagnosis of the source estimate

The counterexample is an exact form computation. Its four checkable inputs are: $A_\varphi = \varphi\nabla\theta$ on the whole open strip; the gauge identity (4); $\int |g_L'|^2 = 2/L$; and $\int V_\varepsilon |g_L|^2 = \varepsilon$. No numerical or asymptotic step is used.

The same geometry diagnoses the source derivation. The claimed magnetic improvement depends on a noninteger flux, but no closed curve in S winds around the boundary pole. Therefore the flux is gauge-trivial on this domain. The source's transverse decomposition into modes indexed by $k \in \mathbb{Z}$ is also incompatible with the constant transverse Neumann ground mode after the global gauge removal. Irrespective of how the primed sum in Theorem 3.2.1 is interpreted, Theorem 1 rules out every finite coefficient multiplying $\|V\|_X$ alone.

6 Novelty check, limitations, and reviewer focus

A bounded search was performed on 13 August 2026. The run's registry, solution, attempt, and proof-gap indexes were searched for arXiv:2106.08625, the exact title, "negative eigenvalues magnetic strip," and Aharonov–Bohm Neumann terms. Web and arXiv searches used the exact title and id, the exact "best constant" phrase, and combinations of "magnetic Schrödinger," "strip," "Neumann," and "Aharonov–Bohm." They found the thesis, conference abstracts repeating its estimate, standard waveguide literature such as [2], and later work on poles approaching a boundary, but no paper correcting this estimate or answering its optimal-constant question. This is a bounded search, not a proof of novelty.

The conclusion concerns the standard gauge-covariant magnetic Neumann realization. If "Neumann" were instead intended to mean the ordinary, non-gauge-covariant condition $\partial_\nu u = 0$ while $A \cdot \nu \neq 0$, that is not the standard magnetic Neumann problem and the source does not give a precise self-adjoint realization; the present packet makes no claim for that different boundary problem. The packet also does not address a pole lying strictly inside the strip, a punctured or multiply connected waveguide, Dirichlet/mixed boundary conditions, curved guides, or estimates with an additive term or stronger norms.

Review should focus on the domain/gauge convention at the boundary pole and on the relative-compactness step for the bounded compactly supported potential. The trial support lies in $\{x \geq 1\}$, so it avoids every local domain subtlety at $(0, 0)$.

References

[1] B. Sorowen, *Estimation of the number of negative eigenvalues of magnetic Schrödinger operators in a strip*, arXiv:2106.08625v1 (2021).

[2] D. Borisov, T. Ekholm, and H. Kovařík, *Spectrum of the magnetic Schrödinger operator in a waveguide with combined boundary conditions*, Annales Henri Poincaré **6** (2005), 327–342; arXiv:math-ph/0405034.